U202115281
Yi Zhenyu

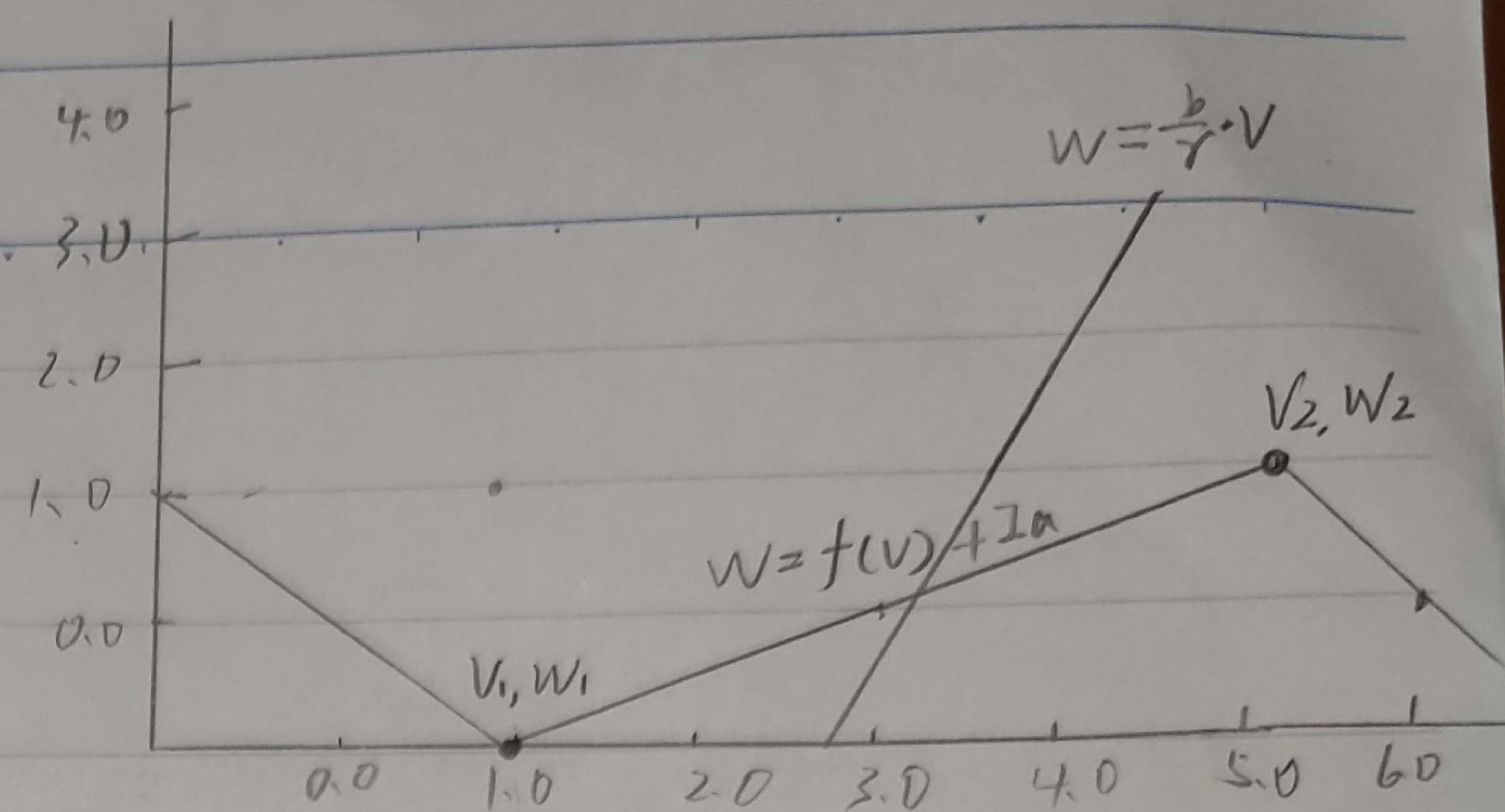
No.

Date

Homework-3

1. Answer:

Oscillatory regime can be gained under the conditions below:



① $\begin{cases} I_1 < I_a < I_2 \\ \tan\theta = \frac{W_2-W_1}{V_2-V_1} < \frac{b}{\gamma} = \frac{b}{1} = b \end{cases}$

And I_i、W_i、V_i can be gained:

② the local minimum/maxmum of V-nullcline,

or we can observe from the curve: $V_1 = 1$, $V_2 = 5$

And $w_i = f(V_i) + I_a$

③ $f(V_i) + I_i = bV_i/\gamma\ (\gamma=1)$ ⇐ the cross of w-nullcline and V-nullcline

Take ②.③ to ① => $\begin{cases} b+1 < I_a < 5b-1 \\ b > 0.5 \end{cases}$

2. Answer:

① For A, $I_A = \int_{y=B}^{y=T} y^2 dA(y) = \int_{y=B}^{y=T} y^2 \cdot w\,dy = \int_{-5}^{-1} y^2 \cdot 4 \cdot dy + \int_{-1}^{1} y^2 \cdot 12\,dy + \int_{1}^{5} y^2 \cdot 4\,dy = \frac{1016}{3}\ cm^2 \approx 338.7\ cm^2$

② For B, $I_A' = \int_{y=B}^{y=T} y^2 dA'(y) = 2\times\left(\int_{-6}^{-4} y^2 dy + \int_{-2}^{2} y^2 dy + \int_{4}^{6} y^2 dy\right) + 10\times\left(\int_{-4}^{-2} y^2 dy + \int_{2}^{4} y^2 dy\right) = \frac{1760}{3}\ cm^2 \approx 586.7\ cm^2$

The area moment of inertia for B is bigger, so B will be more efficient to carry height and resist the bending better.